

Nộp bài tập homework ss12

Tên	Hoàng Duy Hương
Lớp	PD16

- Sau khi restoreDesign HW yêu cầu report time design đúng stage
- Dùng lệnh: timeDesign -postRoute -slackReports (để đúng với stage PostRoute hiện tại)

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timeDesign Summary
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Setup views included:
func_view_wc

-----+-----+-----+-----+-----+-----+-----+-----+-----+
| Setup mode | all | reg2reg | in2reg | reg2out | in2out | default | mem2reg | reg2mem |
-----+-----+-----+-----+-----+-----+-----+-----+-----+
| WNS (ns):  | -2.316 | -0.272 | 0.989 | -2.316 | N/A | 6.793 | 0.008 | 0.200 |
| TNS (ns):  | -132.817 | -104.282 | 0.000 | -28.535 | N/A | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 1307 | 1272 | 0 | 35 | N/A | 0 | 0 | 0 |
| All Paths:  | 10625 | 10058 | 261 | 35 | N/A | 2 | 1265 | 278 |
-----+-----+-----+-----+-----+-----+-----+-----+-----+

-----+-----+-----+-----+
| DRVs | | Real | | Total |
-----+-----+-----+-----+
| | | Nr nets(terms) | Worst Vio | Nr nets(terms) |
-----+-----+-----+-----+
| max_cap | | 128 (128) | -0.027 | 167 (167) |
| max_tran | | 1 (3) | -0.610 | 40 (81) |
| max_fanout | | 1 (1) | -1 | 174 (174) |
| max_length | | 0 (0) | 0 | 0 (0) |
-----+-----+-----+-----+

Density: 45.644%
Total number of glitch violations: 0
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Reported timing to dir ./timingReports
Total CPU time: 43.29 sec
Total Real time: 43.0 sec
Total Memory Usage: 1435.414062 Mbytes
Reset AAE Options
0

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- Đây là summary report timing của postRoute sau khi report.
 - Ở path timing reg2reg thì có 1272 path violate, và có WNS là -0.272 ns
 - Density hiện tại là 45.644% (Ulti ổn)

- Em sẽ tiến hành fix timing setup path đầu tiên của path_group reg2reg này.

Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/CLK
 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/D (*) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/Q (*) triggered by leading edge of 'clk_sys'

Path Groups: {reg2reg}

Analysis View: func_vview_wc
 Other End Arrival Time 1.295
 - Setup 0.198
 + Phase Shift 10.000
 + CPFR Adjustment 0.017
 - Uncertainty 0.100
 = Required Time 11.014
 - Arrival Time 11.287
 = Slack Time -0.272

Clock Rise Edge 0.000
 + Clock Network Latency (Prop) 1.219
 = Beginpoint Arrival Time 1.219

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/CLK	-	28	0.266	0.272	-	1.219	(1179.36, 643.44)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/Q ->	sg13g2_dfrbp_1	1	0.011	0.091	0.462	1.681	(1188.96, 644.28)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP7471_i_ibex_id_stage_i_controller_i_instr_i_17/Y	sg13g2_inv_4	2	0.032	0.060	0.086	1.767	(1190.40, 645.96)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP7616_i_ibex_id_stage_i_controller_i_instr_i_17/Y ->	sg13g2_inv_8	7	0.055	0.060	0.069	1.836	(1196.64, 644.28)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8072_0/Y	sg13g2_nor2b_2	3	0.017	0.131	0.215	2.051	(1228.80, 623.28)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP7735_FE_OFN986_0994/X	sg13g2_buf_8	8	0.065	0.056	0.199	2.313	(1235.52, 604.80)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP7737_FE_OFN986_0994/X	sg13g2_buf_8	8	0.064	0.083	0.282	2.729	(1261.44, 532.56)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_RC_1451_0/Y	sg13g2_a21oi_1	1	0.009	0.142	0.151	2.880	(1336.88, 521.64)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8723_0/Y	sg13g2_a21oi_2	1	0.011	0.140	0.180	3.060	(1334.88, 545.58)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_341_0/Y	sg13g2_nor2b_2	1	0.020	0.088	0.120	3.180	(1331.04, 577.92)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8673_0/Y	sg13g2_nand3_1	1	0.005	0.074	0.088	3.268	(1252.80, 582.12)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OFC5501_i_ibex_ex_block_i_multdiv_operand_a_i_11/X	sg13g2_buf_4	1	0.057	0.112	0.210	3.479	(1256.64, 585.48)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_nand2b_2	1	0.025	0.159	0.193	3.672	(1173.60, 859.32)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4077_0/Y	sg13g2_nand2b_2	1	0.016	0.090	0.121	3.793	(1075.20, 884.52)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_a21oi_2	1	0.011	0.091	0.123	3.916	(1019.52, 885.78)
i_croc_soc/i_croc/i_core_wrap/FE_OCP5254_1065/Y	sg13g2_inv_4	3	0.046	0.091	0.104	4.020	(1022.88, 886.20)
i_croc_soc/i_croc/i_core_wrap/FE_RC_3643_0/Y	sg13g2_nor2_2	1	0.006	0.040	0.074	4.094	(936.48, 834.96)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y	sg13g2_a21oi_2	2	0.012	0.116	0.120	4.214	(936.48, 832.86)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y	sg13g2_inv_2	1	0.007	0.043	0.068	4.282	(935.52, 827.40)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y	sg13g2_nor3_2	2	0.008	0.137	0.132	4.414	(935.52, 819.84)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y	sg13g2_a21oi_2	2	0.018	0.124	0.115	4.529	(934.08, 817.74)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1676_0/Y	sg13g2_nor3_2	1	0.004	0.103	0.159	4.688	(940.32, 804.72)
i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y	sg13g2_nor2b_2	1	0.011	0.106	0.214	4.903	(936.48, 797.16)
i_croc_soc/i_croc/i_core_wrap/FE_OCP6106_FE_RN_1487_0/Y	sg13g2_inv_4	6	0.042	0.078	0.100	5.003	(939.36, 797.16)
i_croc_soc/i_croc/i_core_wrap/FE_RC_185_0/Y	sg13g2_a21oi_2	4	0.018	0.161	0.171	5.174	(894.24, 795.06)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y	sg13g2_a21oi_1	1	0.006	0.108	0.150	5.324	(890.40, 774.48)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1275_0/Y	sg13g2_nand2_2	2	0.007	0.053	0.075	5.399	(886.08, 771.12)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8412_0/Y	sg13g2_nand3_1	1	0.009	0.168	0.184	5.583	(885.60, 768.60)

- Sau khi upsize các cell cần thiết thì path timin đã MET

Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/CLK
 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/D (*) checked with leading edge of 'clk_sys'
 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/Q (*) triggered by leading edge of 'clk_sys'

Path Groups: {reg2reg}

Analysis View: func_vview_wc
 Other End Arrival Time 1.295
 - Setup 0.198
 + Phase Shift 10.000
 + CPFR Adjustment 0.017
 - Uncertainty 0.100
 = Required Time 11.014
 - Arrival Time 11.005
 = Slack Time 0.009

Clock Rise Edge 0.000
 + Clock Network Latency (Prop) 1.219
 = Beginpoint Arrival Time 1.219

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/CLK	-	28	0.266	0.272	-	1.219	(1179.36, 643.44)
i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_17__reg/Q ->	sg13g2_dfrbp_1	1	0.011	0.091	0.462	1.681	(1188.96, 644.28)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP7471_i_ibex_id_stage_i_controller_i_instr_i_17/Y	sg13g2_inv_4	2	0.032	0.060	0.086	1.767	(1190.40, 645.96)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OCP7616_i_ibex_id_stage_i_controller_i_instr_i_17/Y ->	sg13g2_inv_8	7	0.055	0.060	0.069	1.836	(1196.64, 644.28)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8072_0/Y	sg13g2_nor2b_2	3	0.031	0.216	0.278	2.115	(1228.80, 623.28)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP7735_FE_OFN986_0994/X	sg13g2_buf_16	8	0.065	0.056	0.199	2.313	(1235.52, 604.80)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OCP7737_FE_OFN986_0994/X	sg13g2_buf_8	8	0.064	0.070	0.154	2.467	(1261.44, 532.56)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_RC_1451_0/Y	sg13g2_a22oi_1	1	0.009	0.142	0.146	2.612	(1336.88, 521.64)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8723_0/Y	sg13g2_a21oi_2	1	0.011	0.140	0.180	2.792	(1334.88, 545.58)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_341_0/Y	sg13g2_nor2b_2	1	0.020	0.088	0.123	2.915	(1331.04, 577.92)
i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8673_0/Y	sg13g2_nand3_1	1	0.009	0.100	0.111	3.026	(1252.80, 582.12)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OFC5501_i_ibex_ex_block_i_multdiv_operand_a_i_11/X	sg13g2_buf_8	1	0.057	0.068	0.161	3.186	(1258.08, 585.48)
i_croc_soc/i_croc/i_core_wrap/FE_RC_724_0/Y	sg13g2_nand2b_2	1	0.025	0.155	0.187	3.373	(1173.60, 859.32)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4077_0/Y	sg13g2_nand2b_2	1	0.016	0.089	0.123	3.496	(1075.20, 884.52)
i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_a21oi_2	1	0.011	0.091	0.123	3.619	(1019.52, 885.78)
i_croc_soc/i_croc/i_core_wrap/FE_OCP5254_1065/Y	sg13g2_inv_4	3	0.046	0.091	0.118	3.737	(1022.88, 886.20)
i_croc_soc/i_croc/i_core_wrap/FE_RC_3643_0/Y	sg13g2_nor2_2	1	0.006	0.040	0.075	3.812	(936.48, 834.96)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y	sg13g2_a21oi_2	2	0.012	0.116	0.120	3.932	(936.48, 832.86)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y	sg13g2_inv_2	1	0.007	0.043	0.068	4.000	(935.52, 827.40)
i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y	sg13g2_nor3_2	2	0.008	0.137	0.132	4.132	(935.52, 819.84)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y	sg13g2_a21oi_2	2	0.018	0.124	0.115	4.248	(934.08, 817.74)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1676_0/Y	sg13g2_nor3_2	1	0.004	0.103	0.159	4.407	(940.32, 804.72)
i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y	sg13g2_nor2b_2	1	0.011	0.106	0.214	4.621	(936.48, 797.16)
i_croc_soc/i_croc/i_core_wrap/FE_OCP6106_FE_RN_1487_0/Y	sg13g2_inv_4	6	0.042	0.078	0.100	4.721	(939.36, 797.16)
i_croc_soc/i_croc/i_core_wrap/FE_RC_185_0/Y	sg13g2_a21oi_2	4	0.018	0.161	0.171	4.892	(894.24, 795.06)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y	sg13g2_a21oi_1	1	0.006	0.108	0.150	5.042	(890.40, 774.48)
i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1275_0/Y	sg13g2_nand2_2	2	0.007	0.053	0.075	5.117	(886.08, 771.12)

- Report lại summary timing

timeDesign Summary									
Setup views included: func_view_wc									
Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	mem2reg	reg2mem	
WNS (ns):	-2.316	-0.024	0.989	-2.316	N/A	6.793	0.008	0.452	
TNS (ns):	-28.589	-0.054	0.000	-28.535	N/A	0.000	0.000	0.000	
Violating Paths:	42	7	0	35	N/A	0	0	0	
All Paths:	10625	10058	261	35	N/A	2	1265	278	

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	128 (128)	-0.027	167 (167)
max_tran	1 (3)	-0.610	40 (81)
max_fanout	1 (1)	-1	174 (174)
max_length	0 (0)	0	0 (0)

Density: 45.648%

Total number of glitch violations: 0

Reported timing to dir ./timingReports

Total CPU time: 39.57 sec

Total Real time: 39.0 sec

Total Memory Usage: 1505.71875 Mbytes

Reset AAE Options

0

- Path setup violate thứ 2

0 Path 1: VIOLATED Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_0_70__reg/CLK									
9 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_0_70__reg/0 (*) checked with leading edge of 'clk_sys'									
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_19__reg/Q (*) triggered by leading edge of 'clk_sys'									
11 Path Groups: {reg2reg}									
12 Analysis View: func_view_wc									
13 Other End Arrival Time									
14 - Setup	1.295								
15 + Phase Shift	0.198								
16 + CPFR Adjustment	10.000								
17 - Uncertainty	0.017								
18 = Required Time	0.100								
19 - Arrival Time	11.014								
20 = Slack Time	11.038								
	-0.024								
21 Clock Rise Edge	0.000								
22 + Clock Network Latency (Prop)	1.315								
23 = Beginpoint Arrival Time	1.315								
Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location		
28 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_19__reg/CLK	-	32	0.168	0.149	-	1.315	(1153.44, 774.90)		
29 i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_i_controller_i_instr_i_19__reg/Q ->	sg13g2_dfrbp_2	6	0.029	0.118	0.499	1.814	(1160.16, 774.48)		
30 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/ictc_postCTS_setupFE_0FC5157_i_ibex_id_stage_i_controller_i_instr_i_19/Y	sg13g2_inv_2	3	0.016	0.065	0.098	1.912	(1214.88, 782.04)		
31 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/4108_0/Y	sg13g2_nand2_1	1	0.009	0.002	0.091	2.003	(1216.80, 787.50)		
32 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/ictc_postCTS_setupFE_0CPC7611_1066/X ->	sg13g2_buf_8	7	0.087	0.093	0.168	2.171	(1215.36, 789.60)		
33 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/ictc_postCTS_setupFE_0FC4682_1066/X	sg13g2_buf_2	8	0.088	0.295	0.344	2.515	(1438.56, 689.64)		
34 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_8723_0/Y	sg13g2_a21o1_2	1	0.011	0.111	0.158	2.674	(1334.88, 545.58)		
35 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_341_0/Y	sg13g2_nor2b_2	1	0.020	0.166	0.194	2.867	(1331.04, 577.92)		
36 i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_i/FE_RC_0672_0/Y	sg13g2_nand2_1	1	0.009	0.100	0.214	3.081	(1252.00, 582.12)		
37 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_0FC5501_i_ibex_ex_block_i_multdiv_operand_a_i_11/X	sg13g2_buf_8	1	0.053	0.065	0.201	3.282	(1258.08, 585.48)		
38 i_croc_soc/i_croc/i_core_wrap/FE_RC_724_0/Y	sg13g2_nand2b_2	1	0.029	0.113	0.128	3.410	(1173.60, 859.32)		
39 i_croc_soc/i_croc/i_core_wrap/FE_RC_4077_0/Y	sg13g2_nand2b_2	1	0.016	0.111	0.144	3.554	(1075.20, 884.52)		
40 i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_a21o1_2	1	0.011	0.115	0.151	3.705	(1019.52, 885.78)		
41 i_croc_soc/i_croc/i_core_wrap/FE_0CPC5254_1065/Y	sg13g2_inv_4	3	0.045	0.083	0.112	3.816	(1022.88, 886.20)		
42 i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y	sg13g2_a21o1_2	2	0.012	0.116	0.149	3.965	(936.48, 832.80)		
43 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y	sg13g2_inv_2	1	0.007	0.043	0.068	4.033	(935.52, 827.40)		
44 i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y	sg13g2_nor3_2	2	0.008	0.137	0.132	4.165	(935.52, 819.84)		
45 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y	sg13g2_a21o1_2	2	0.018	0.124	0.115	4.280	(934.08, 817.74)		
46 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1076_0/Y	sg13g2_nor3_2	1	0.004	0.103	0.159	4.440	(940.32, 804.72)		
47 i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y	sg13g2_nor2b_2	1	0.011	0.106	0.214	4.654	(936.48, 797.16)		
48 i_croc_soc/i_croc/i_core_wrap/FE_0CPC6106_FE_RN_1487_0/Y	sg13g2_inv_4	6	0.042	0.078	0.100	4.754	(939.36, 797.16)		
49 i_croc_soc/i_croc/i_core_wrap/FE_RC_185_0/Y	sg13g2_a21o1_2	4	0.018	0.161	0.171	4.925	(894.24, 795.00)		
50 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y	sg13g2_a21o1_1	1	0.006	0.108	0.150	5.015	(890.40, 774.48)		
51 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1275_0/Y	sg13g2_nand2_2	2	0.007	0.053	0.075	5.150	(886.08, 771.12)		
52 i_croc_soc/i_croc/i_core_wrap/FE_RC_8412_0/Y	sg13g2_nand3_1	1	0.009	0.168	0.183	5.333	(885.60, 768.60)		
53 i_croc_soc/i_croc/i_core_wrap/FE_RC_8620_0/Y	sg13g2_a21o1_2	2	0.009	0.103	0.158	5.491	(983.84, 767.34)		
54 i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1740_0/Y	sg13g2_a21o1_2	2	0.028	0.169	0.204	5.696	(907.68, 767.34)		

- Đã MET path setup thứ 2

8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/CLK							
9 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/D (*) checked with leading edge of 'clk_sys'							
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_19__reg/Q (*) triggered by leading edge of 'clk_sys'							
11 Path Groups: (reg2reg)							
12 Analysis View: func_view_wc							
13 Other End Arrival Time 1.295							
14 - Setup 0.198							
15 + Phase Shift 10.000							
16 + CPFR Adjustment 0.017							
17 - Uncertainty 0.100							
18 = Required Time 11.014							
19 - Arrival Time 10.836							
20 = Slack Time 0.179							
21 Clock Rise Edge 0.000							
22 + Clock Network Latency (Prop) 1.315							
23 = Beginpoint Arrival Time 1.315							

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location

27							
28	i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_19__reg/CLK	-	32	0.168	0.149	-	1.315 (1153.44, 774.90)
29	i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_19__reg/Q ->	sg13g2_dfrbp_2	6	0.029	0.118	0.499	1.814 (1160.16, 774.48)
30	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OFC3157_i_ibex_id_stage_1_controller_1_instr_1_19/Y	sg13g2_inv_2	3	0.016	0.065	0.098	1.912 (1214.38, 782.04)
31	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/4108/Y	sg13g2_nand2_1	1	0.009	0.082	0.091	2.003 (1216.80, 787.50)
32	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OPCPT611_1066/X ->	sg13g2_buf_8	7	0.093	0.098	0.171	2.174 (1215.36, 789.60)
33	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OFC4682_1066/X ->	sg13g2_buf_8	8	0.088	0.095	0.194	2.367 (1440.00, 689.64)
34	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8723_0/Y	sg13g2_a210i_2	1	0.011	0.094	0.103	2.471 (1334.88, 545.58)
35	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_341_0/Y	sg13g2_nor2b_2	1	0.020	0.164	0.193	2.664 (1331.04, 577.92)
36	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8673_0/Y	sg13g2_nand3_1	1	0.009	0.179	0.213	2.877 (1252.80, 582.12)
37	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OFC5501_i_ibex_ex_block_1_multdiv_operand_a_1_11/X	sg13g2_buf_8	1	0.053	0.065	0.201	3.077 (1258.08, 585.48)
38	i_croc_soc/i_croc/i_core_wrap/FE_RC_724_0/Y	sg13g2_nand2b_2	1	0.029	0.113	0.129	3.206 (1173.60, 859.32)
39	i_croc_soc/i_croc/i_core_wrap/FE_RC_4077_0/Y	sg13g2_nand2b_2	1	0.016	0.111	0.144	3.351 (1075.20, 884.52)
40	i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_a210i_2	1	0.011	0.115	0.151	3.501 (1010.52, 885.76)
41	i_croc_soc/i_croc/i_core_wrap/FE_OPC5254_1065/Y	sg13g2_inv_4	3	0.045	0.083	0.113	3.614 (1022.88, 886.20)
42	i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y	sg13g2_a210i_2	2	0.012	0.116	0.149	3.763 (936.48, 832.86)
43	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y	sg13g2_inv_2	1	0.007	0.043	0.068	3.831 (935.52, 827.40)
44	i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y	sg13g2_nor_2	2	0.008	0.137	0.132	3.063 (935.52, 819.84)
45	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y	sg13g2_a210i_2	2	0.018	0.124	0.115	4.078 (934.08, 817.74)
46	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1676_0/Y	sg13g2_nor3_2	1	0.004	0.103	0.159	4.238 (940.32, 804.72)
47	i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y	sg13g2_nor2b_2	1	0.011	0.106	0.214	4.452 (936.48, 797.16)
48	i_croc_soc/i_croc/i_core_wrap/FE_OPC6106_FE_RN_1487_0/Y	sg13g2_inv_4	6	0.042	0.078	0.100	4.552 (939.36, 797.16)
49	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y	sg13g2_a210i_2	4	0.010	0.161	0.171	4.723 (894.24, 795.06)
50	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1000_0/Y	sg13g2_a21a1_1	1	0.006	0.108	0.150	4.673 (890.40, 774.48)
51	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1275_0/Y	sg13g2_nand2_2	2	0.007	0.053	0.075	4.948 (886.08, 771.12)
52	i_croc_soc/i_croc/i_core_wrap/FE_RC_8412_0/Y	sg13g2_nand3_1	1	0.009	0.168	0.183	5.131 (885.60, 768.60)
53	i_croc_soc/i_croc/i_core_wrap/FE_RC_8620_0/Y	sg13g2_a210i_2	2	0.009	0.103	0.158	5.289 (903.84, 767.34)
54	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1740_0/Y	sg13g2_a210i_2	2	0.028	0.169	0.204	5.493 (907.68, 767.34)

- Fix tiếp path thứ 3

8 Path 1: MET Setup Check with Pin i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/CLK							
9 Endpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_crash_dump_o_70__reg/D (*) checked with leading edge of 'clk_sys'							
10 Beginpoint: i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q (*) triggered by leading edge of 'clk_sys'							
11 Path Groups: (reg2reg)							
12 Analysis View: func_view_wc							
13 Other End Arrival Time 1.295							
14 - Setup 0.198							
15 + Phase Shift 10.000							
16 + CPFR Adjustment 0.017							
17 - Uncertainty 0.100							
18 = Required Time 11.014							
19 - Arrival Time 10.988							
20 = Slack Time 0.026							
21 Clock Rise Edge 0.000							
22 + Clock Network Latency (Prop) 1.219							
23 = Beginpoint Arrival Time 1.219							

Pin	Cell	Fanout	Load	Slew	Delay	Arrival Time	Pin Location

27							
28	i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/CLK	-	28	0.266	0.272	-	1.219 (1190.88, 635.88)
29	i_croc_soc/i_croc/i_core_wrap/i_ibex_id_stage_1_controller_1_instr_1_15__reg/Q ->	sg13g2_dfrbp_1	1	0.022	0.152	0.500	1.725 (1200.48, 636.72)
30	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OPC9154_i_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_8	5	0.073	0.082	0.108	1.833 (1202.40, 636.72)
31	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OPC9159_i_ibex_id_stage_1_controller_1_instr_1_15/Y	sg13g2_inv_16	7	0.147	0.085	0.088	1.921 (1250.40, 665.70)
32	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OPC9164_i_ibex_id_stage_1_controller_1_instr_1_15/Y ->	sg13g2_inv_16	8	0.121	0.067	0.102	2.023 (1339.20, 692.58)
33	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OPC9197_0/Y	sg13g2_inv_2	2	0.012	0.050	0.095	2.119 (1373.28, 530.88)
34	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_OFC9463_FE_OCPN6487_FE_OFN573_i_ibex_id_stage_1_controller_1_instr_1_15/X	sg13g2_buf_8	8	0.037	0.050	0.128	2.246 (1376.16, 532.56)
35	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_2729_0/Y	sg13g2_nand2_1	1	0.004	0.067	0.084	2.330 (1343.52, 517.86)
36	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_2727_0/Y	sg13g2_a21a1_1	1	0.004	0.122	0.070	2.408 (1339.68, 523.32)
37	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/ictc_postCTS_setupFE_RC_1451_0/Y	sg13g2_a210i_1	1	0.009	0.142	0.155	2.562 (1336.80, 521.64)
38	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8723_0/Y	sg13g2_a210i_2	1	0.011	0.116	0.189	2.742 (1334.88, 545.58)
39	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_341_0/Y	sg13g2_nor2b_2	1	0.020	0.082	0.111	2.854 (1331.04, 577.92)
40	i_croc_soc/i_croc/i_core_wrap/i_ibex_register_file_1/FE_RC_8673_0/Y	sg13g2_nand3_1	1	0.005	0.074	0.085	2.889 (1252.80, 582.12)
41	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_OFC5501_i_ibex_ex_block_1_multdiv_operand_a_1_11/X	sg13g2_buf_4	1	0.057	0.112	0.210	3.149 (1256.64, 585.48)
42	i_croc_soc/i_croc/i_core_wrap/FE_RC_724_0/Y	sg13g2_nand2b_2	1	0.025	0.159	0.205	3.354 (1173.60, 859.32)
43	i_croc_soc/i_croc/i_core_wrap/FE_RC_4077_0/Y	sg13g2_nand2b_2	1	0.016	0.090	0.124	3.478 (1075.20, 884.52)
44	i_croc_soc/i_croc/i_core_wrap/FE_RC_4076_0/Y	sg13g2_a210i_2	1	0.011	0.091	0.123	3.601 (1019.52, 885.76)
45	i_croc_soc/i_croc/i_core_wrap/FE_OPC5254_1065/Y	sg13g2_inv_4	3	0.046	0.091	0.119	3.720 (1022.88, 886.20)
46	i_croc_soc/i_croc/i_core_wrap/FE_RC_3643_0/Y	sg13g2_nor_2	1	0.006	0.040	0.070	3.795 (936.48, 834.96)
47	i_croc_soc/i_croc/i_core_wrap/FE_RC_8768_0/Y	sg13g2_a210i_2	2	0.012	0.116	0.129	3.915 (936.48, 832.86)
48	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_199_0_dup/Y	sg13g2_inv_2	1	0.007	0.043	0.068	3.983 (935.52, 827.40)
49	i_croc_soc/i_croc/i_core_wrap/FE_RC_8441_0/Y	sg13g2_nor3_2	2	0.008	0.137	0.132	4.115 (935.52, 819.84)
50	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1491_0/Y	sg13g2_a210i_2	2	0.018	0.124	0.115	4.230 (934.08, 817.74)
51	i_croc_soc/i_croc/i_core_wrap/ictc_postCTS_setupFE_RC_1676_0/Y	sg13g2_nor3_2	1	0.004	0.103	0.150	4.390 (940.32, 804.72)
52	i_croc_soc/i_croc/i_core_wrap/FE_RC_3627_0/Y	sg13g2_nor2b_2	1	0.011	0.106	0.214	4.604 (936.48, 797.16)
53	i_croc_soc/i_croc/i_core_wrap/FE_OPC6106_FE_RN_1487_0/Y	sg13g2_inv_4	6	0.042	0.078	0.100	4.704 (939.36, 797.16)

- Report lại summary timing

timeDesign Summary									
Setup views included: func_view_wc									
Setup mode	all	reg2reg	in2reg	reg2out	in2out	default	mem2reg	reg2mem	
WNS (ns):	-2.316	-0.007	0.989	2.316	N/A	6.793	0.008	0.469	
TNS (ns):	-28.542	-0.007	0.000	-28.535	N/A	0.000	0.000	0.000	
Violating Paths:	36	1	0	35	N/A	0	0	0	
All Paths:	10625	10058	261	35	N/A	2	1255	278	

DRVs	Real		Total
	Nr nets (terms)	Worst Vio	Nr nets (terms)
max_cap	128 (128)	-0.027	167 (167)
max_tran	1 (3)	-0.610	40 (81)
max_fanout	1 (1)	-1	174 (174)
max_length	0 (0)	0	0 (0)

Density: 45.646%

Total number of glitch violations: 0

Reported timing to dir ./timingReports

Total CPU time: 39.73 sec

Total Real time: 40.0 sec

Total Memory Usage: 1373.796875 Mbytes

Reset AAE Options

0

- Fix DRC/short/overlap
 - Fix overlap

```

innovus 53> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=1648.2M, init mem=1667.2M)
Overlapping with other instance:      2
*info: Placed = 50308                (Fixed = 9472)
*info: Unplaced = 0
Placement Density:45.65%(631397/1383097)
Placement Density (including fixed std cells):46.31%(648300/1400000)
Finished checkPlace (total: cpu=0:00:01.5, real=0:00:01.0; vio checks: cpu=0:00:01.0, real=0:00:01.0; mem=1681.2M)
1
innovus 54> win
innovus 55> checkPlace
**Info: (IMPSP-307): Design contains fractional 1 cell.
Begin checking placement ... (start mem=1685.3M, init mem=1685.3M)
*info: Placed = 50308                (Fixed = 9471)
*info: Unplaced = 0
Placement Density:45.65%(631402/1383101)
Placement Density (including fixed std cells):46.31%(648300/1399998)
Finished checkPlace (total: cpu=0:00:01.6, real=0:00:01.0; vio checks: cpu=0:00:01.1, real=0:00:01.0; mem=1685.3M)
0
innovus 56>

```

- Check DRC

- Không hiểu sao fix nhiều path timing tool có thể phát sinh thêm DRC nên path này trở lên khá nhiều lỗi. cách fix nhanh là route lại ở stage này thì DRC sẽ mất. Nhưng em sẽ restore lại design này lại và fix DRC thủ công!

```
Verification Complete : 260 Viols.

Violation Summary By Layer and Type:

      Mar   Short  MetSpc  CutSpc  CShort  Totals
Metal1    97    73    48      0      0    218
Vial       0      0      0      5      4      9
Metal2     0    18      9      0      0    27
Metal3     0      0      2      0      0      2
Metal4     0      0      3      0      0      3
Metal5     0      1      0      0      0      1
Totals    97    92    62      5      4    260

*** End Verify DRC (CPU: 0:00:10.6  ELAPSED TIME: 11.00  MEM: 0.0M) ***

1
innovus 57> █
```

- Sau khi restore lại

```
Verification Complete : 13 Viols.

Violation Summary By Layer and Type:

      MetSpc   Mar   Short  NSMet  Totals
Metal1        2     3     1      1      7
Metal3        2     0     0      0      2
Metal4        3     0     0      0      3
Metal5        0     0     1      0      1
Totals        7     3     2      1     13

*** End Verify DRC (CPU: 0:00:10.9  ELAPSED TIME: 11.00  MEM: 0.0M) ***

1
innovus 57> █
```

Violation Type:

- Verify (13/13)
 - Geometry (13/13)
 - Metal_Short (2/2)
 - Metal1(1) (1/1)
 - Metal5(5) (1/1)
 - Minimal_Area (3/3)
 - Metal1(1) (3/3)
 - NSMetal (1/1)
 - Metal1(1) (1/1)
 - Parallel_Run_Length_Spacing (7/7)
 - Metal1(1) (2/2)
 - Metal3(3) (2/2)
 - Metal4(4) (3/3)

Violation:

LAYER	OBJECT1	OBJECT2
Metal1(1)	NET i_croc_soc/i_croc/i_cor...	
Metal1(1)	NET i_croc_soc/i_croc/i_cor...	INST i_croc_soc
Metal1(1)	NET i_croc_soc/i_croc/i_cor...	
Metal1(1)	NET i_croc_soc/i_croc/i_cor...	INST i_croc_soc
Metal1(1)	NET i_croc_soc/i_croc/i_cor...	INST i_croc_soc
Metal1(1)	NET i_croc_soc/i_croc/i_tim...	
Metal1(1)	NET i_croc_soc/i_croc/i_tim...	
Metal3(3)	NET i_croc_soc/i_croc/i_cor...	NET i_croc_soc
Metal3(3)	NET i_croc_soc/i_croc/i_cor...	NET i_croc_soc
Metal4(4)	NET i_croc_soc/i_croc/i_cor...	NET i_croc_soc
Metal4(4)	NET i_croc_soc/i_croc/i_cor...	NET i_croc_soc
Metal4(4)	NET i_croc_soc/i_croc/i_cor...	NET i_croc_soc

- Fix DRC các lỗi short và spacing

```
VERIFY DRC ..... Sub-Area : 80 complete 0 Viols.  
VERIFY DRC ..... Sub-Area: {1457.920 1457.920 1640.000 1640.000} 81 of 81  
VERIFY DRC ..... Sub-Area : 81 complete 0 Viols.  
  
Verification Complete : 0 Viols.  
  
*** End Verify DRC (CPU: 0:00:10.9 ELAPSED TIME: 10.00 MEM: 0.0M) ***  
  
1
```

- Đã clean DRC!

```
17:32:42 *** PROCESSED 45000 NETS.  
  
Begin Summary  
  6 Problem(s) (IMPVFC-94): The net has dangling wire(s).  
  6 total info(s) created.  
End Summary  
  
End Time: Wed Aug 19 17:32:42 2026  
Time Elapsed: 0:00:02.0  
  
***** End: VERIFY CONNECTIVITY *****  
Verification Complete : 6 Viols.  0 Wrngs.  
(CPU Time: 0:00:01.5 MEM: 0.000M)  
  
1  
innovus_70>
```

- Còn 6 lỗi connect

Violation Type:

- [-] Verify (8/8)
 - [-] Connectivity (6/6)
 - [-] Antenna (DanglingWire) (2/2)
 - Metal1(1) (2/2)**
 - Open (2/2)
 - [-] UnConnPin (2/2)
 - Metal1(1) (2/2)
 - [-] Geometry (2/2)
 - [-] Minimal_Area (2/2)
 - Metal1(1) (2/2)

Violation:

LAYER	OBJECT1	LOCATION
Metal1(1)	i_croc_soc/i_croc/i_timer/FE_...	(578.16, 952.14) ...
Metal1(1)	i_croc_soc/i_croc/i_timer/co...	(577.44, 953.4) (...)

Description:

- Fix connect

```
**** 14:52:26 **** Processed 45000 nets.  
  
Begin Summary  
  Found no problems or warnings.  
End Summary  
  
End Time: Thu Aug 20 14:52:26 2026  
Time Elapsed: 0:00:02.0  
  
***** End: VERIFY CONNECTIVITY *****  
Verification Complete : 0 Viols. 0 Wrngs.  
(CPU Time: 0:00:01.5 MEM: 0.000M)  
  
1
```

- Đã clean!